

PARKTOWN BOYS' HIGH SCHOOL



Preliminary Examination Paper 1

30 August 2024

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner

Mrs S Rowell

Moderator

Ms H Gouws

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This paper consists of 8 pages, an answer sheet and a formula sheet.
 - You may detach the formula sheet for your own use.
 - Detach the answer sheet, write your name on it and hand it in with your answers.
2. There are 10 questions. Answer all the questions.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
5. Answers only will NOT necessarily be awarded full marks.
6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
7. If necessary, round off answers to TWO decimal places, unless stated otherwise.
8. Diagrams are NOT necessarily drawn to scale.
9. Use the mark allocation to guide the length of your solution.
10. Write neatly and legibly.
11. Rule off after each question.

QUESTION 1

1.1 Solve for x :

1.1.1 $x(x+4) = 3(x+4)$ (3)

1.1.2 $6x^2 - 3x = 11$ (correct to TWO decimal places). (3)

1.1.3 $\frac{6}{x+3} - \frac{x}{1-x} = \frac{8x-4}{x^2+2x-3}$ (5)

1.1.4 $-x^2 + 3x > -10$ (4)

1.2 Solve simultaneously for x and y if:

$x + y = 3$ and $2x^2 - y^2 = 7$ (5)

1.3 Given: $A = (1-x)$ and $B = (1+x)(1+x^2)(1+x^4)\dots(1+x^{256})$
Determine the value of $A \times B$ in terms of x . (3)

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QUESTION 2

2.1 Show that the geometric series $27 - 9 + 3 - 1 + \dots$ converges and calculate the sum of the series. (5)

2.2 Determine the sum of the geometric series $126 + 42 + 14 + \dots + \frac{14}{729}$. (5)

[10]

QUESTION 3

3.1 Given the quadratic sequence: $3; -3; -7; m; \dots$

3.1.1 Write down the value of m . (1)

3.1.2 Determine the n^{th} term of this sequence. (3)

3.1.3 The first difference between two consecutive terms of the sequence is 54.
Calculate the value of these two terms. (4)

3.2 The sum of the fifth term and the eighth term of an arithmetic sequence is 50.
The sum of the first eight terms is 136. Determine the first three terms of the sequence. (4)

3.3 If $\sum_{k=2}^{11} 2 \cdot \sqrt{3^k} = x(\sqrt{3} + 1)$, determine the value of x . (5)

[17]

QUESTION 4

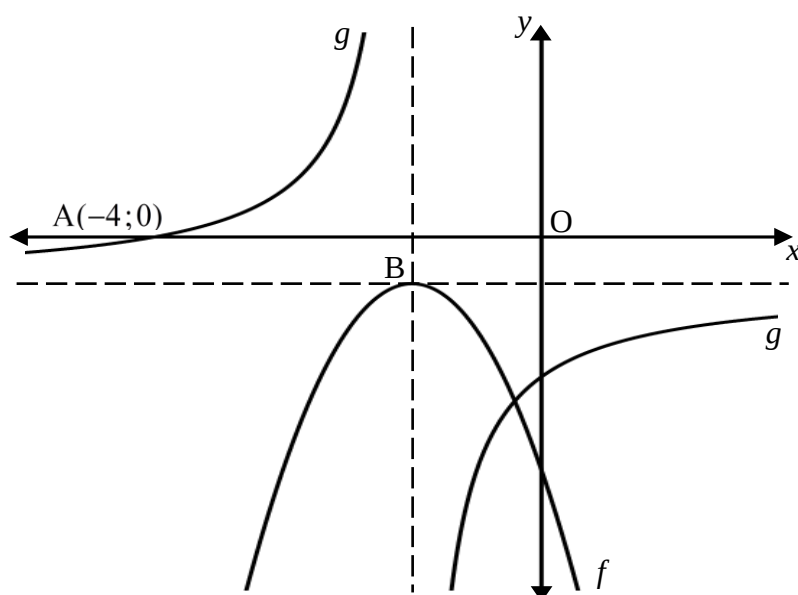
4.1 Given: $f(x) = -2^x + 4$

4.1.1 Write down the equation of the asymptote of f . (1)

4.1.2 Sketch the graph of f on the answer sheet provided, clearly showing all relevant details. **Remember to detach the answer sheet and hand it in with your answer book.** (4)

4.1.3 $h(x) = -2^{x+a} + b$ is obtained by shifting f horizontally and vertically. The coordinates of the x -intercept of h is $(-2; 0)$ and the range of h is $y \in (-\infty; 1)$. Determine the values of a and b . (2)

4.2 Sketched below is the function $f(x) = -x^2 - 4x - 5$ and the hyperbola g , with equation $(x+p)(y+q) = -2t$. $A(-4; 0)$ is the x -intercept of g . f has a turning point at B which also intersects the asymptotes of g .



4.2.1 Determine the coordinates of B . (3)

4.2.2 Write down the domain of g . (1)

4.2.3 Determine the equation of the line of symmetry of g with a negative gradient. (2)

4.2.4 Determine the equation of the tangent of f at $x = -\frac{1}{2}$. (4)

4.2.5 Determine the values p , q and t . (5)

4.2.6 Write down the values of x for which $f'(x) \cdot g(x) \geq 0$. (3)

[25]

QUESTION 5

Given: $f(x) = -2\sqrt{x}$

- 5.1 Determine the equation of f^{-1} in the form $y = \dots$ (2)
- 5.2 Sketch the graphs of f and f^{-1} on the answer sheet provided and show the line of symmetry. **Remember to detach the answer sheet and hand it in with your answer book.** (5)
- 5.3 Determine the coordinates of the point of intersection of f and f^{-1} . (2)
- [9]**

QUESTION 6

- 6.1 The nominal interest rate of an investment is 12,35% p.a., compounded quarterly. Calculate the effective interest rate. (3)
- 6.2 How many years did it take for the cost of a loaf of bread to increase from R11,43 to R17,68, if the average inflation rate is 5,6%? (3)
- 6.3 Thomas takes out a loan of R120 000. He is to repay the loan by means of equal monthly payments, starting one month after the loan was granted. The loan is to be repaid over a period of 4 years, at an interest rate of 8,75% p.a. compounded monthly.
- 6.3.1 What is the monthly repayment? (3)
- 6.3.2 Thomas is unable to make the 9th, 10th and 11th payment. He still wants to pay off the loan by the end of the original 4 years. What will his new monthly payment be? (5)
- [14]**

QUESTION 7

7.1 If $f(x) = -x^3$, determine $f'(x)$ from first principles. (5)

7.2 Find $\frac{dy}{dx}$ for each of the following:

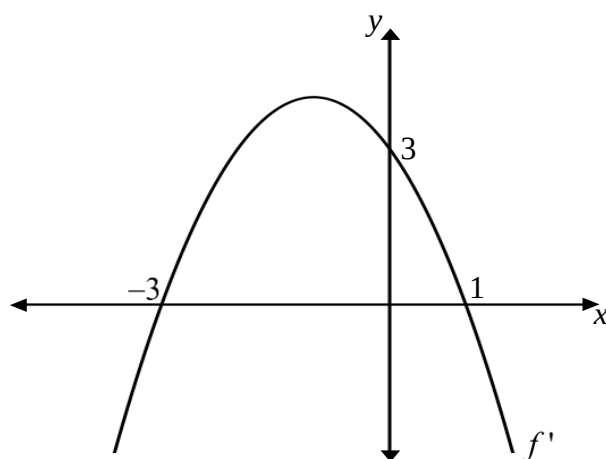
7.2.1 $y = \frac{3}{x^2} - 10\sqrt[5]{x}$ (3)

7.2.2 $y = \frac{x^3 + 1}{x + 1}$ (3)

[11]

QUESTION 8

The sketch below shows the graph of f' .



8.1 What is the gradient of the tangent of f at $x = 0$? (1)

8.2 Write down the x-coordinates of the stationary points of f . (2)

8.3 What is the gradient of the tangent of f at $x = -2$? (1)

8.4 For which values of x is f

8.4.1 increasing? (2)

8.4.2 concave down? (2)

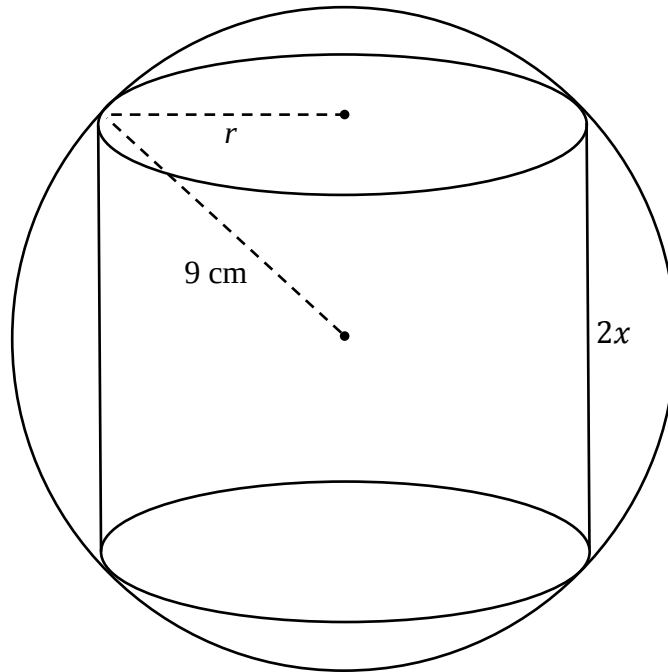
8.5 Draw a rough sketch of f if $f(0) = -2$. (4)

8.6 Hence, or otherwise, determine the equation of f . (5)

[17]

QUESTION 9

A cylinder with height $2x$ cm and a radius r cm is placed inside a sphere with a radius of 9 cm.



9.1 Show that the volume of the cylinder is $V(x) = 162\pi x - 2\pi x^3$. (3)

9.2 Calculate the maximum volume of the cylinder. (5)

[8]

QUESTION 10

- 10.1 A and B are two events in a sample space where $P(\text{not } A) = 0,32$ and $P(B) = 0,17$.

Determine $P(A \text{ or } B)$ if A and B are independent events. (3)

- 10.2 A rack has 7 black blazers and 9 red blazers. Two blazers are randomly removed and given to learners in need.

10.2.1 Represent this scenario by means of a tree diagram. (4)

10.2.2 Calculate the probability that both blazers were red. (1)

10.2.3 Calculate the probability that a red and a black blazer were removed. (2)

- 10.3 A random 7 digit password is generated, using the digits 1 to 7.

10.3.1 If digits may repeat, what is the probability that the password will End with an even digit? (3)

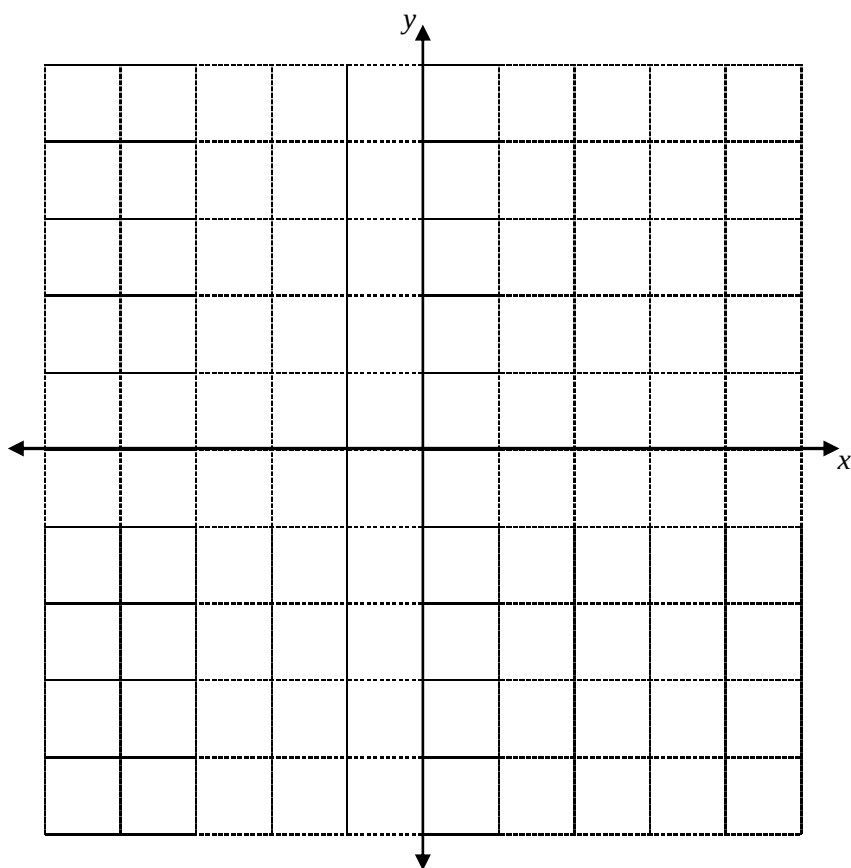
10.3.2 If digits may not repeat, what is the probability that 6 and 7 will not appear next to each other in the password? (3)

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QUESTION 4.1.2



QUESTION 5.2

